MILITARY SPECIFICATION

REGULATOR, OXYGEN

This specification is approved for use within the Defense Personnel Support Center and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a single-stage pressure reducing oxygen regulator suitable for use in the aeromedical evacuation system aboard C-141B aircraft.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-RST, 2800 S. 20th Street, Philadelphia, PA 19101, by using self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6680
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SPECIFICATIONS

FEDERAL

PPP-B-601 - Boxes, Wood, Cleated-plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-L-10547 - Liners, Case and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads.
MIL-STD-794 - Parts and Equipment, Procedures for Packaging of.

(Copies of specifications and standards, required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B57.1 Compressed Gas Cylinder Valve Outlet and Inlet Connections.

(Application for copies should be addressed to the Publication Sales Dept., STD, American Institute of Physics, 335 E. 45th Street, NY, NY 10017).

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COMPRESSED GAS ASSOCIATION, INC. (CGA)

CGA Pamphlet V5	-	Diameter - Index Safety System (Medical Applications); Non-Interchangeable Low Pressure Connections For Medical Gases, Air and Suction.
CGA E-7	-	Standard for Flowmeters, Pressure Reducing Regulators, Regulator/Flowgauge Combination For the Administration of Medical Gases.

(Application for copies should be addressed to Compressed Gas Association, Inc., 1235 Jefferson Davis Highway, Arlington, VA 22202).

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Fire and casualty hazards.

3.1.1 The contractor shall submit proof, prior to delivery, that the items to be delivered under this specification conform to the applicable requirements of the CGA Standard E-7.

The proof shall consist of a laboratory report from a nationally recognized independent laboratory adequately equipped and competent to determine compliance with the CGA Standard specified above. The government reserves the right to require a facilities report and to conduct a facilities survey. Any laboratory report submitted shall include actual test methods, data taken, and results of tests. Detailed findings of examinations necessary to determine compliance shall also be included. The test report shall also have a section that describes the test sample in such a way that the report can be used to insure identity of production items with the test sample. Compliance with these requirements does not excuse compliance with all other requirements of the contract and specifications.

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3.2 Suitability for use in military aeromedical evacuation environment. The product to be delivered shall comply with all applicable requirements of the Test and Evacuation Planning Guide for Aeromedical Evacuation Equipment. Prior to award of the contract, contractor shall submit a first article sample, complete assembly drawing and bill of material. The first article sample shall be considered acceptable only upon approval by the Air Force, for use in the aeromedical evacuation system.

3.3 Material. The materials used in the construction of the regulator shall be in accordance with the materials specified in Table I. The body of the regulator shall be fabricated of machined copper alloy, chromium plated or chromium plated over nickel-plated. The copper alloy used shall have a minimum tensile strength of 50,000 pounds per square inch. Any material not specified shall be of the highest quality, suitable in all respect for the purpose intended.

NOTE: If internal plating exists, flaking or other visible indication of poor plating adhesion shall be considered a defective regulator.

3.4 Style, design and dimensions. The regulator shall consist essentially of a union nut and gland, a body, a pressure control valve for manually controlling the outlet pressure, an outlet pressure gage, and a standard externally-threaded fitting for connection to low-pressure oxygen devices. The weight of the unit shall not exceed three pounds. Style, design, and dimensions of the regulator shall be in accordance with Figure 1.

NOTE: Any variation in material, and internal construction may be permissible if the regulator meets all other requirements of this specification and is approved by the Air Force as specified in 3.2.

3.4.1 Regulator. Regulator shall be of a single stage type which shall maintain a minimum discharge rate of 160 liters per minute. The pressure of the outlet flow shall be manually adjustable by means of a pressure control valve in the range of 0 to 100 pounds per square inch gage (psig). Turning the knob in clockwise direction shall increase the pressure. Counter-clockwise rotation of the knob shall decrease the pressure. The oxygen flow shall not be reduced below 160 LPM when the inlet pressure varies in the range of 2000 to 300 psig. A suitable safety relief valve shall be incorporated to prevent accumulation of hazardous pressure within the regulator. The pressure relief valve shall be fully activated at 125 psig and shall automatically reseat when the pressure returns to 100 psig or below. The regulator shall be equipped with a commercial union nut and gland inlet connection which shall be capable of making connection with suitable oxygen cylinder having a 0.903-14 NGO right hand thread on the outlet valve

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in accordance with CGA-540. The regulator inlet shall have a suitable, assembled filter with openings not greater than 60 microns or equivalent mesh, fabricated from bronze to prevent foreign matter from entering the regulator.

On the high-pressure side of the regulator, an internally threaded outlet connection shall be provided which shall be suitable for making leakproof connection with a standard high pressure gage. A suitable plug shall be provided to close this outlet when the high pressure gage is not connected. This plug shall make tight, leakproof connection with the regulator body.

3.4.2 Low-pressure oxygen fitting. On the low-pressure side of the regulator, a standard threaded oxygen fitting shall be provided for low-pressure application, in accordance with CGA connection No. 1240, as specified in CGA Pamphlet V5. This standard fitting shall make tight, leakproof connection with the regulator body.

3.4.3 Pressure gage. The regulator shall be equipped with a pressure gage on the low-pressure side suitable for pressure measurement from 0 to 100 psig. The gage shall be two inches in diameter and 1-1/8 inches deep. The gage shall be calibrated to indicate pressure. The gage shall be marked in 2 psig increments from 0-100 psig, with red marking at each 10 psig point. The markings shall be numbered at every 20 psig point. Dial of the gage will be white. The face cover of the gage shall be fabricated from clear and durable plastic.

3.4.4 Flow rate accuracy. When tested as specified in 4.4.1, the actual flow delivery by the regulator shall be a minimum of 160 LPM. The regulator shall maintain this accuracy for any inlet pressure between 300 and 2000 psig and for any outlet pressure between 50 to 100 psig.

3.4.5 Regulator leakage. When tested as specified in 4.4.2, the regulator shall not exhibit leakage as evidence by formation of bubbles.

3.5 Thread sealants. Organic-based thread sealants which are considered to be flammable when in contact with oxygen shall not be used. Thread sealants used for antiseize and sealing purposes shall be polytetrafluoroethylene thread-sealing tape or thread compounds. Thread sealing tape shall be applied so that it is not permitted to enter the oxygen flow stream. Thread sealing tape shall be applied with at least one internal male thread uncovered.

3.6 Service data. The contractor shall enclose, with each unit, two copies each of one or more booklets which contain complete instructions for installations, operation, maintenance, and a list of component parts.

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3.7 Markings. All high pressure and low pressure fittings on the regulator body shall be appropriately identified with permanent marking. The pressure control valve knob shall be clearly marked identifying direction of increasing and decreasing pressure. The regulator body shall be clearly and legibly marked with the manufacturer's name or registered trademark and part number.

3.8 Workmanship. Regulators shall be free from defects which detract from their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and as indicated in table I. The unit of product for examination purpose shall be one regulator.

TABLE I. Sampling for examination.

Type of defect	Inspection level	AQL (percent defective)
Major	II	1.0
Minor	II	2.5

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4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in table II. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

TABLE II. Sampling for tests

Characteristic	Requirement	Test procedure	Inspection level	AQL (percent defective)
Dimensions	3.4	---	S-3	0.65
Flow rate accurency	3.4.4	4.4.1	S-3	1.0
Leakage test	3.4.5	4.4.2	S-3	1.0

4.3 Examination.

4.3.1 Examination of regulators. The regulators shall be examined to determine compliance with all the requirements contained in this specification. Examination shall be conducted as specified in tables I and III. Examination, however, shall not be restricted to defects listed in table III. Any regulators in the sample containing one or more defects shall be rejected and if the number of defective regulators in any sample exceeds the acceptance number for the sample, the lot represented by that sample shall be rejected.

TABLE III. Classification of defects

Categories	Defects
Critical	None defined
Major	
101	External brass surfaces not plated.
102	Overall design not as specified.
103	External surfaces not free from corrosion, scale, pits, holes and crevices.
104	Regulator gage calibration markings not as specified.
105	Complete unit weight exceeds three pounds.
106	Pressure control valve not easily adjustable.
107	One or more components loose.
108	Face cover of pressure gage not fabricated from clear plastic.
109	Regulator inlet does not possess filter.
110	Regulator inlet or outlet blocked with foreign matter.
111	Service data not furnished as specified.
112	Face cover not free of scratches or cracks.
Minor	
201	Markings not as specified.

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4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Flow rate accuracy test. This test shall be performed in carefully controlled surroundings to insure maximum operator safety. A pneumatic circuit shall be assembled which shall consist, essentially, of the regulator to be tested (sample), a standard oxygen, compressed nitrogen, or compressed air cylinder with at least 90 percent of full charge remaining, an adjustable calibrated regulator and pressure gage with at least 3000 psig capacity, a calibrated precision flow-readout device to measure actual flow output of the sample, and necessary delivery tubing.

The adjustable, calibrated regulator shall be inserted between the pressure source and the sample. The outlet of the sample shall be connected to the calibrated flowmeter. All connections shall be made in a leakproof manner.

The test shall be conducted by regulating the pressure source to 2000, 1500, 1000, 500, and 300 psig with the sample regulator pressure adjusting knob set at 50, 75 and 100 psig respectively. Each test shall consist of recording the actual output flows of the sample in LPM by regulating the pressure source with the calibrated, adjustable regulator, in descending values; and subsequently recording the actual output flows of the sample by regulating the pressure source in ascending values. This procedure will result in obtaining two actual output flow values of the sample for each of the above regulated pressures and sample pressure setting.

4.4.2 Leakage test. The leakage test shall be performed, essentially, by utilizing the same circuit specified in 4.4.1, except that the precision flowmeter may be removed from the circuit. The oxygen cylinder shall be capable of delivering at least 2000 psig pressure to the regulator. The regulator pressure adjusting knob shall be turned out fully (turn counter-clockwise), and a film of oxygen compatible leak detection agent shall be applied to all regulator junctions with 2000 psig internal pressure applied to the regulator. The regulator output shall then be suitably plugged. The regulator pressure-adjusting knob shall be turned all the way in (turn clockwise). All junctions shall be observed for formation of any bubbles. As an alternate method, the entire regulator may be immersed in water at a depth no greater than 6 inches.

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4.5 Inspection of packaging. The sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. Regulator shall be cleaned and dried in accordance with the applicable method as outlined in MIL-P-116.

5.1.1.2 Preservative applications. Preservatives shall not be used.

5.1.1.3 Unit package. Each regulator shall be packaged in accordance with MIL-P-116, method IC-2.

5.1.2 Level C. The level C preservation for regulator shall conform to the MIL-STD-794 requirements for this level.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Six unit packages shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, overseas type, PPP-B-621, class 2, or PPP-B-636, class weather resistant. Closure and strapping shall be as specified in the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall be as specified in the liner specification. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in PPP-B-636.

5.2.2 Level B. Six unit packages shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, domestic type, PPP-B-621, class 1, or PPP-B-636, class domestic. Closure shall be as specified in the applicable box specification.

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5.2.3 Level C. The level C packing for regulator shall be packed in accordance with the requirements of MIL-STD-794 for level C.

5.2.4 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in a exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.2.5 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Loads shall be unitized on pallets conforming to MIL-STD-147. Quantities less than 40 cubic feet need not be unitized. Pallet loads including the pallet shall not exceed 43 inches in height, 48 inches in width, and 40 inches in length, and shall conform to Type IV or Type V.

5.2.5.1 Levels A and B. Regulator packed as specified in 5.2.1 and 5.2.2 shall be unitized on pallets as specified in 5.2.5.

5.2.5.2 Level C. Regulator packed as specified in 5.2.3 shall be unitized as specified in MIL-STD-794.

5.3 Marking.

5.3.1 Levels A, B and C. Each unit pack, intermediate package, exterior container, and unitized load shall be marked as specified in MIL-STD-129.

5.4 General.

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.5.

6. NOTES

6.1 Intended use. The regulator described herein shall be used for the administration of therapeutic oxygen to patients in the aeromedical evacuation system.

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6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(c) Applicable National Stock Number.

6.3 National Stock Number Coverage. This specification covers the following National Stock Number:

National Stock No.	Item Identification
6680-01-183-9784	REGULATOR, OXYGEN, Single Stage Pressure Reducing

6.5 Subject term (key word) listing.

Constant flow oxygen regulator
Oxygen regulator
Pressure reducing regulator
Pressure regulator

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing Activity:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITY:

VA-OSS
PHS
FDA-MPQAS

Project No. 6680-0218

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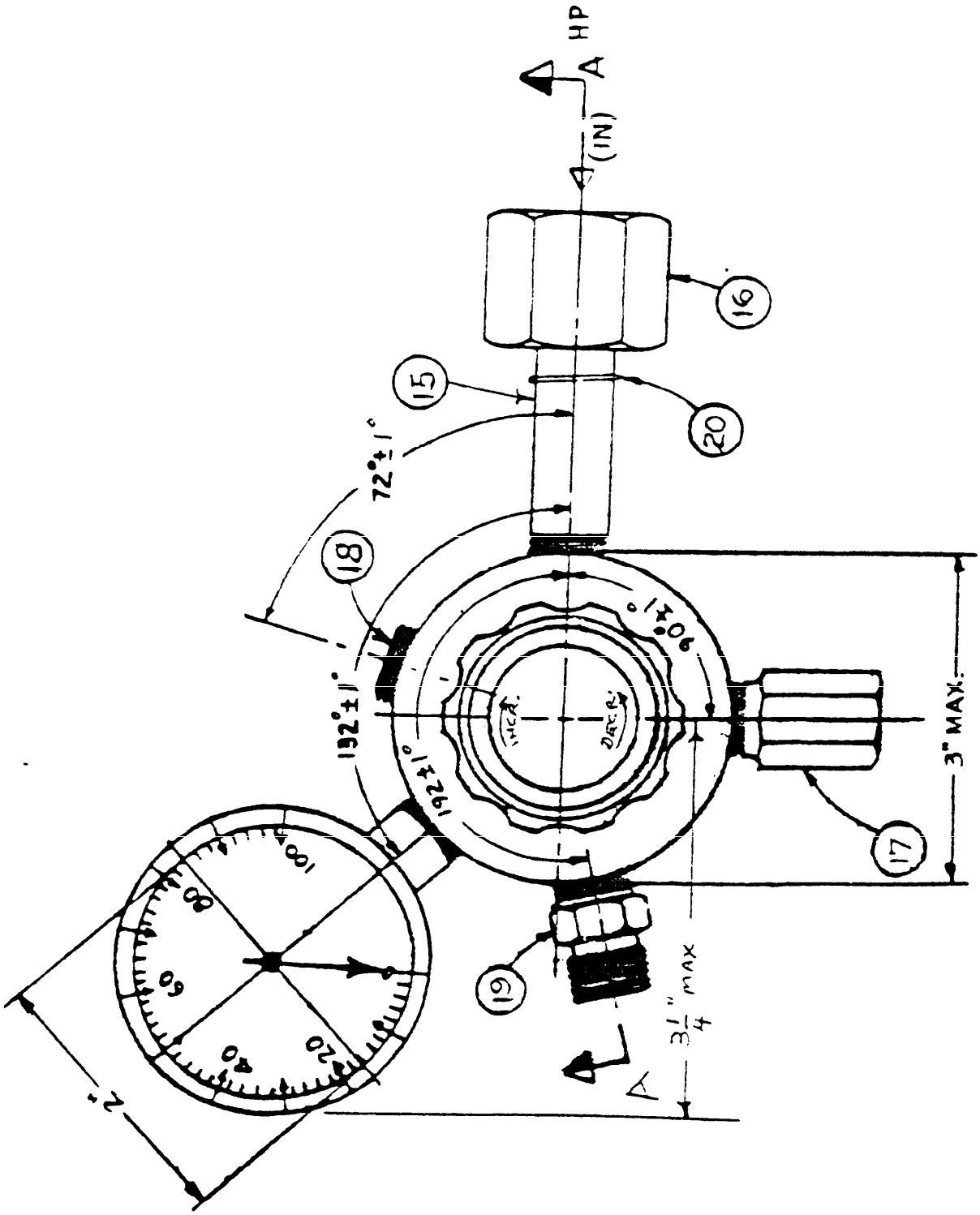


FIGURE 1A REGULATOR, OXYGEN

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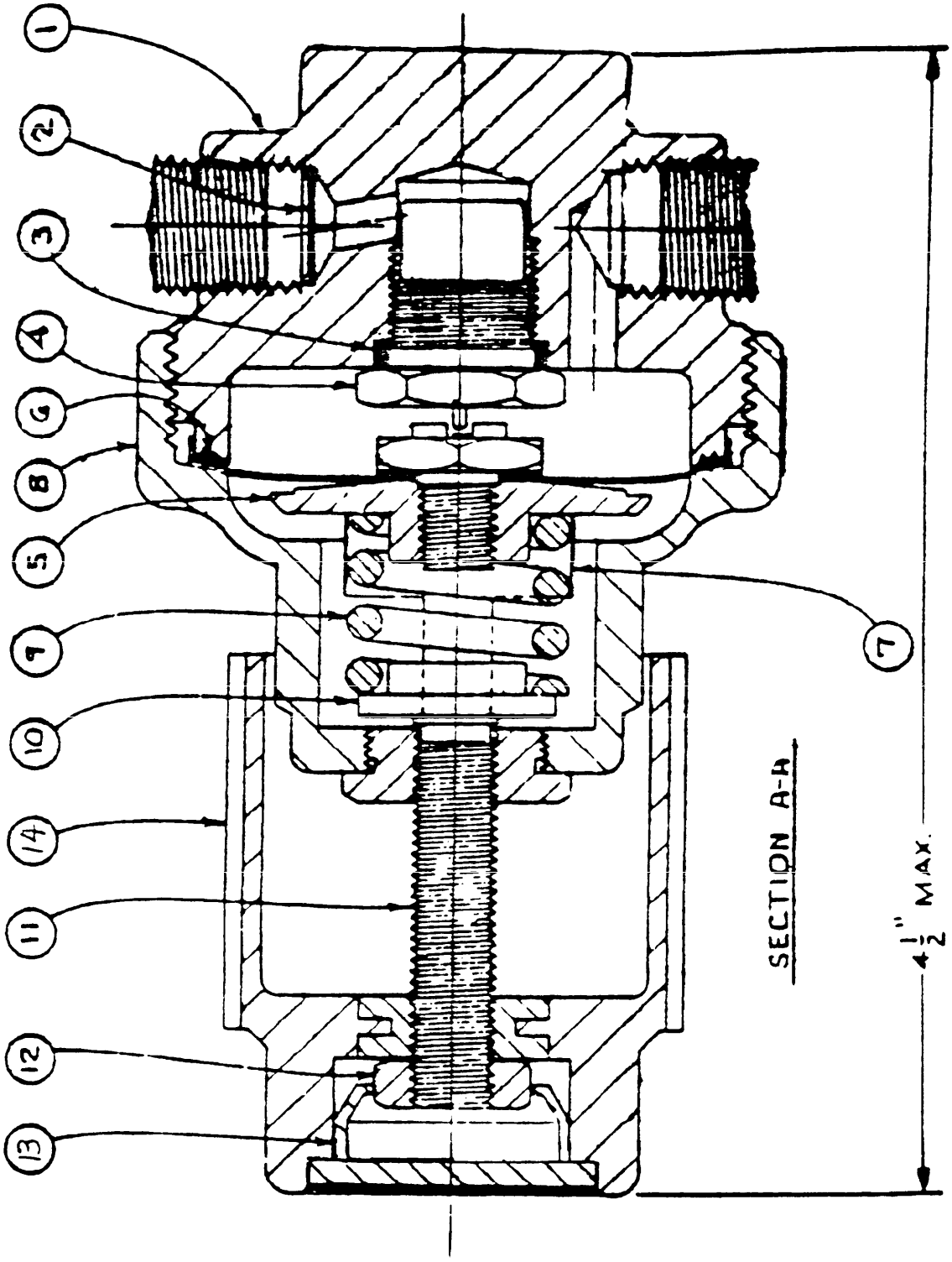


FIGURE 1B REGULATOR, OXYGEN

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ITEM	DESCRIPTION	MATERIAL
1	BODY	BRASS, CHROME, PL.
2	SCREEN	316 CORROSION RESIST. STEEL
3	GASKET	TETRAFLUOROPOLYETHYLENE
4	VALVE SEAT ASSEMBLY	BRASS, FLUORAL POLYMER, 316 CORROSION-RESISTANT STEEL
5	DIAPHRAGM ASSEMBLY	301 CORROSION-RESISTING STEEL
6	GASKET	NYLON
7	BAND, FRICTION	316 CORROSION RESIST STEEL
8	CAP ASSEMBLY	BRASS, CHROME PLATED
9	SPRING	STEEL, ZINC PLATED
10	BUTTON	BRASS, ACETAL
11	STEM	ALUMINUM, BRONZE
12	NUT	STEEL, ZINC PLATED
13	CLEVIS	STEEL, CHROME PLATED
14	KNOB, PRESSURE ADJUSTING	ABS, BRASS
15	NIPPLE ASSEMBLY	BRASS, CHROME PLATED
16	NUT,	BRASS, CHROME PLATED
17	RELIEF VAVE ASSEMBLY	BRASS, NEOPRENE
18	PLUG	BRASS, CHROME PLATED
19	ADAPTER	BRASS, CHROME PLATED
20	RING, RETAINING	STEEL, CHROME PLATED

TABLE - 1

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R 42043(DM)	REGULATOR, OXYGEN

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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